

20260618 Bin by EWHa or sSFR

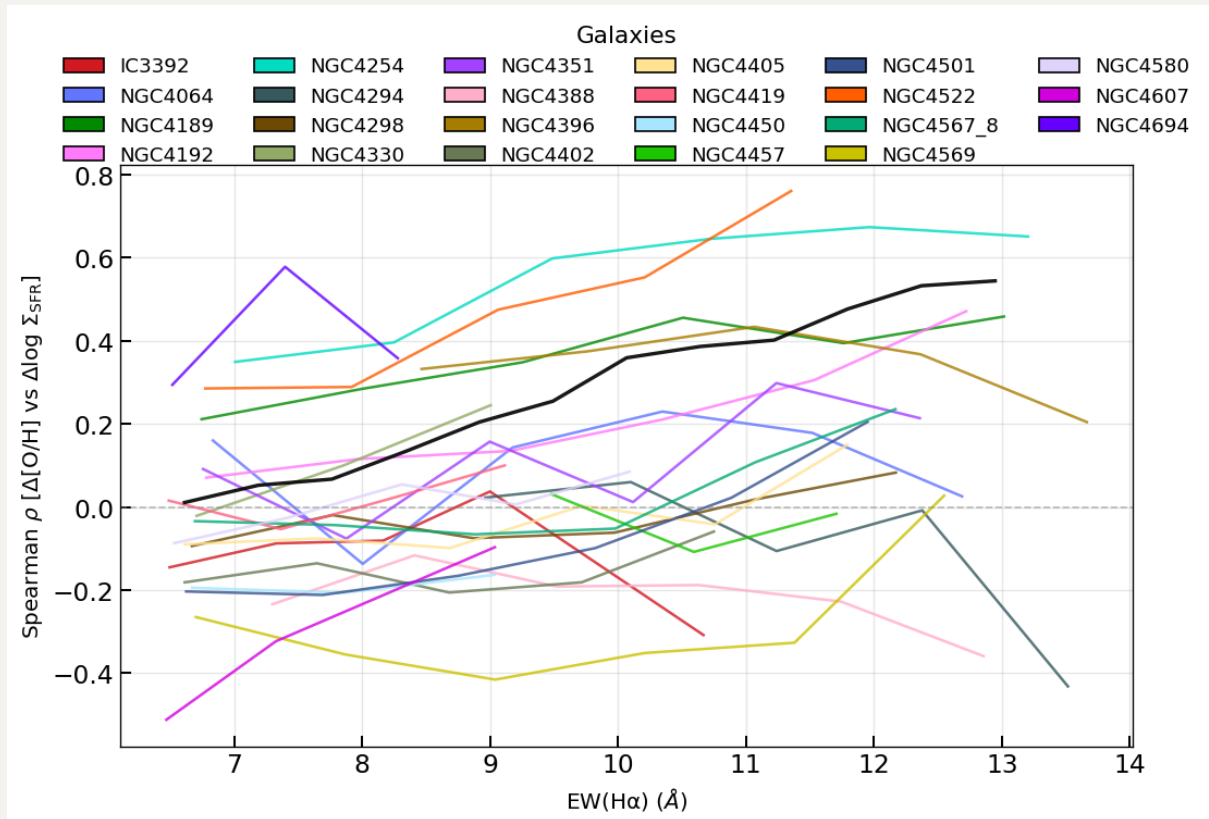
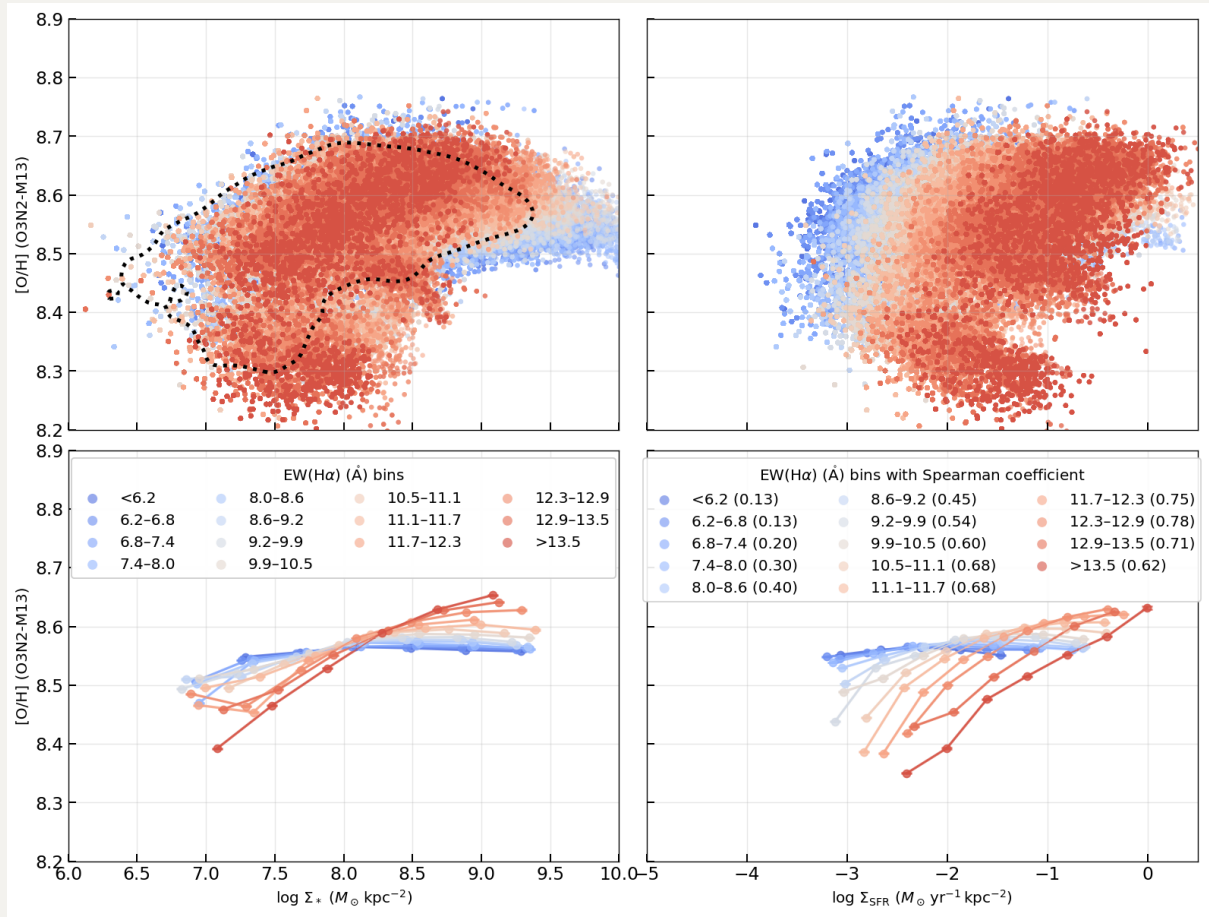
Here, instead of binning by Σ_* , we try to bin by $EW(H\alpha)$ or $sSFR$ to see if we still see the inversion. So like in $rMZR$ and $O/H-SFR$ relations, and when calculating the $\Delta \log_{10}(\Sigma_{SFR})$ and $\Delta[O/H]$ to get the Moran maps and Spearman trends, we now bin by $EW(H\alpha)$ or $sSFR$ first and then do the subtraction

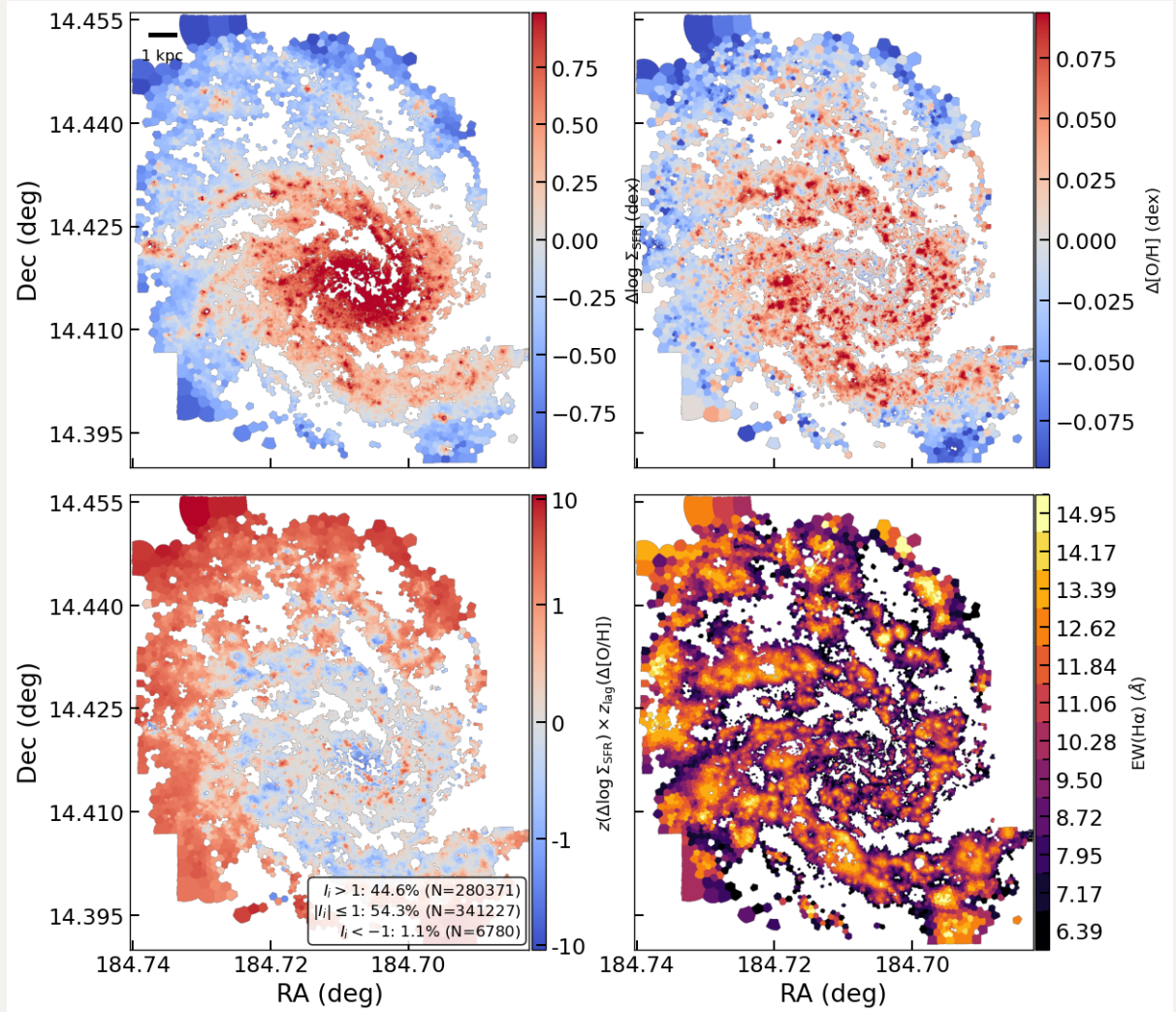
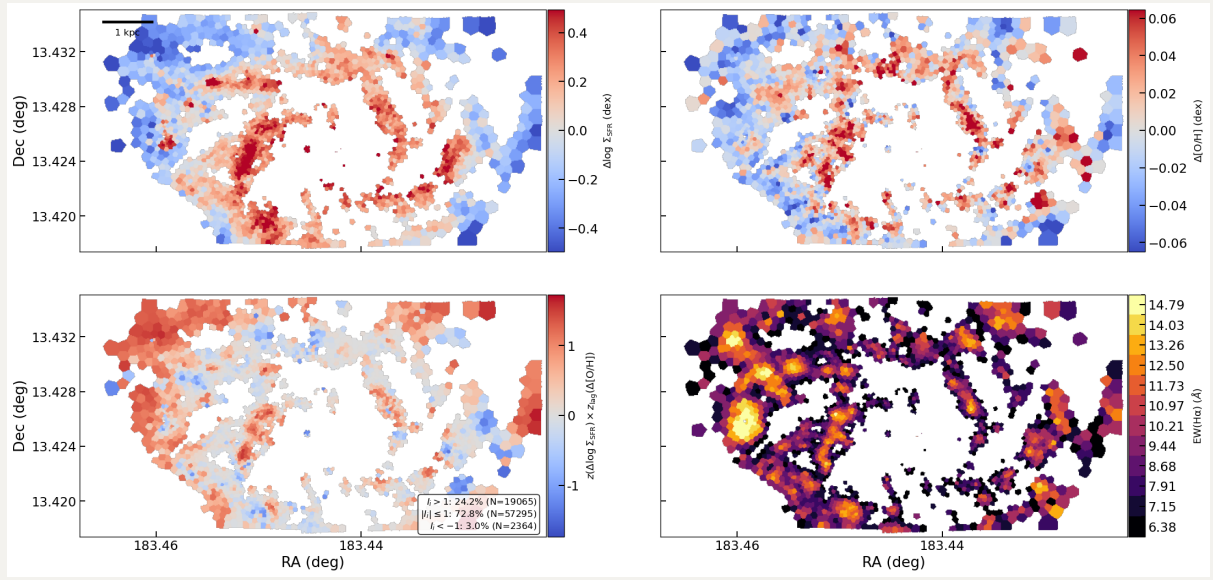
Unfortunately, the results are hard to comment. It seems that in ensemble trend, both binning approaches wash out the reversal. But overall, all galaxies seem to present much more flatten Spearman trends and thus no more internal reversals as well.

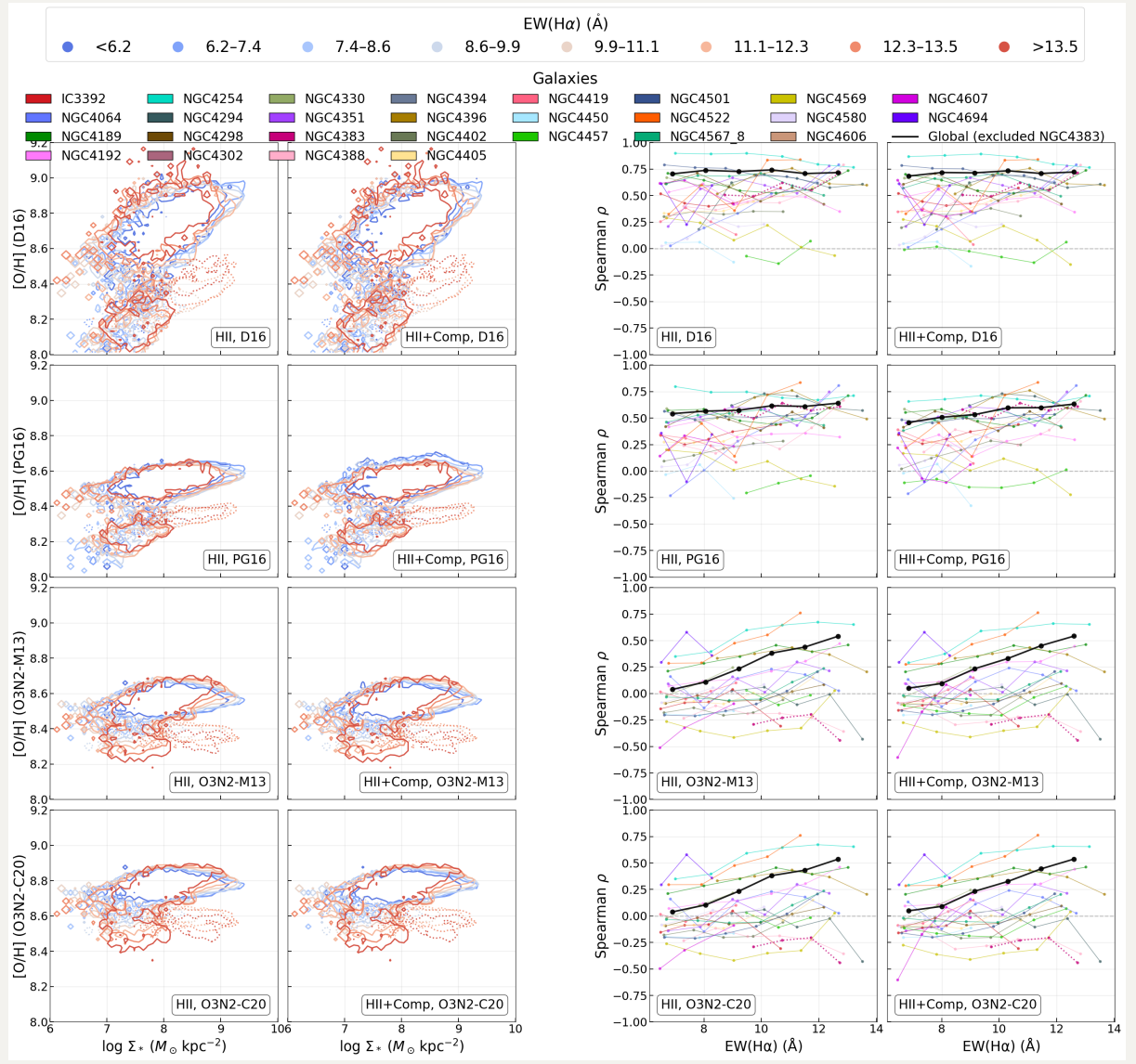
My feeling is that these two may not be a good proxy to replace Σ_* when doing $\Delta \log_{10}(\Sigma_{SFR})$ and $\Delta[O/H]$, but they should be meaningful in some senses that i still have no idea.

As for the replacement of Σ_* , we still need to find other "truely" independent parameter, as both $EW(H\alpha)$ and $sSFR$ are still related to $H\alpha$ -inferred SFR.

Bin by $EW(H\alpha)$







Bin by sSFR

